

PROJECT CHARTER

Enterprise Aviation Data Platform (EADP)

1. Project Overview

ABC Airways Ltd. operates more than 500 daily domestic and international flights and generates large volumes of data across independent airline systems. The project will design and implement an enterprise aviation data platform that brings heterogeneous operational and passenger data into a governed, analytics-ready environment.

2. Business Problem

The airline's operational information is distributed across independent systems such as flight operations, passenger reservations, airport operations, baggage, maintenance, crew scheduling, ticketing, CRM, flight delay reporting and aircraft IoT monitoring. This separation makes enterprise-wide operational monitoring, passenger analytics, maintenance planning and executive reporting difficult.

3. Project Objective

- Collect airline operational data from multiple heterogeneous enterprise systems.
- Clean, validate and standardize passenger, flight and maintenance data.
- Store curated datasets in an enterprise PostgreSQL data warehouse.
- Maintain metadata, audit information and end-to-end data lineage.
- Support airline operations, passenger analytics, maintenance planning and executive reporting.
- Use Git and GitHub for collaborative version control and project traceability.
- Produce enterprise-standard technical documentation and sprint deliverables.

4. Current Scope

4.1 Sprint 0 — Project Initiation & Architecture

- Business problem and stakeholder analysis
- Business Requirement Document (BRD)
- High-level Solution Architecture
- Project Charter
- Product Backlog and Sprint 0 Backlog
- GitHub repository and enterprise repository structure

4.2 Sprint 1 — Data Discovery & Ingestion

- Source-system analysis and Source Inventory

- Data Dictionary
- Pentaho ingestion transformations for the available source formats
- PostgreSQL staging/raw landing zone
- Logging and exception-handling approach
- Git commits for successful ingestion work

5. Project Boundaries

The current academic assignment scope is limited to Sprint 0 and Sprint 1 execution. Later activities described in the capstone roadmap, such as full data profiling, warehouse implementation, lineage implementation and executive Power BI dashboards, are treated as planned/future backlog work unless the faculty expands the scope.

6. Stakeholders

Stakeholder	Primary Interest / Need
Flight Operations Team	Flight performance, punctuality and delay visibility
Airport Operations Team	Airport activity, facility and operational information
Customer Service Team	Passenger issues, feedback and service performance
Maintenance Team	Aircraft maintenance planning and aircraft condition information
Commercial Team	Booking, ticketing, passenger and revenue-related insights
Executive Management	Enterprise operational and financial reporting
Project Team	Data engineering, documentation, version control and delivery

7. Technology Stack

Category	Technology	Purpose
ETL	Pentaho Data Integration (Spoon)	Data ingestion and transformation
Database	PostgreSQL	Staging and enterprise warehouse storage
Programming	Python / Pandas	Data profiling and quality analysis
Version Control	Git / GitHub	Collaborative development and history
Reporting	Power BI	Analytics and reporting
Documentation	Markdown / MS Word	Technical and project documentation
Methodology	Agile Scrum	Sprint-based project

		execution
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8. Risks and Mitigations

Risk	Impact	Mitigation
Heterogeneous formats	High	Use source-specific Pentaho transformations and standard staging conventions.
Data quality issues	High	Profile, validate and document quality issues before downstream use.
Large datasets	Medium/High	Use efficient ingestion, database loading and controlled test samples.
Schema mismatch across sources	High	Maintain a data dictionary and source-to-target mapping.
Team integration conflicts	Medium	Use Git branches/commits and a shared repository structure.
ETL failures	Medium	Implement logging, validation and exception-handling steps.

9. Success Criteria

- All Sprint 0 deliverables are documented and mutually consistent.
- All 10 selected datasets are mapped to the intended airline source systems.
- Source formats and structures are documented.
- Sprint 1 ingestion pipelines are designed/implemented using Pentaho as required.
- Raw/staging data can be loaded into PostgreSQL.
- GitHub contains traceable project artifacts and commits.